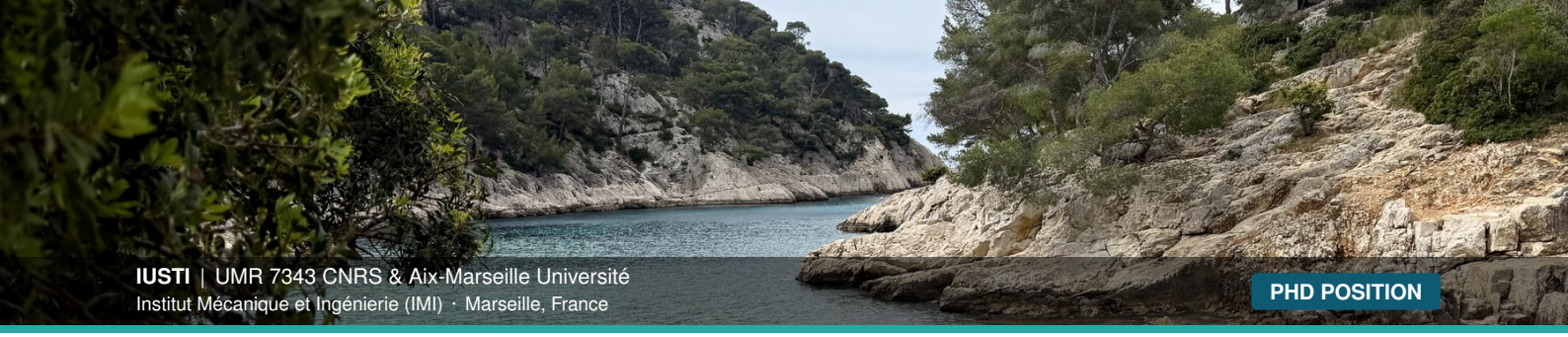


SNSF project *Hierarchy of Scales* · WP 3.2

From network architecture to fracture energy

Linking polymer network structure to dissipation in soft fracture

* * *

POSITION	PhD student, experimental	**DURATION**	3 years (doctoral contract)	**START**	As soon as possible
PARTNERS	ETH Zurich (D. Kammer, simulations) · CNRS Lyon (M. Adda-Bedia, theory) · IUSTI & IMI, Aix-Marseille Université				

The sub-project. The energy required to break a polymer network sets the intrinsic toughness of soft materials, yet even for simple gels the link between network architecture and fracture energy remains debated. The classical Lake–Thomas picture attributes the intrinsic fracture energy to the scission of the chains that cross the crack plane [1], but real networks contain loops, dangling ends, and spatial heterogeneities that change how much energy is stored and dissipated before and around the advancing crack [2]. This PhD project (WP 3.2) will compare model networks with a nearly ideal, homogeneous structure (tetra-PEG gels [3]) with conventional, heterogeneous polyacrylamide gels. Fracture energies will be measured with pure-shear tests [4], with J-integral analysis of measured near-tip fields [5], and from crack-tip opening displacements. Network structure will be imaged directly, in the bulk and close to crack surfaces, using AMU's cryo-electron microscopy facilities. The results will provide experimental benchmarks for phantom-network simulations at ETH Zurich and for theory in Lyon.

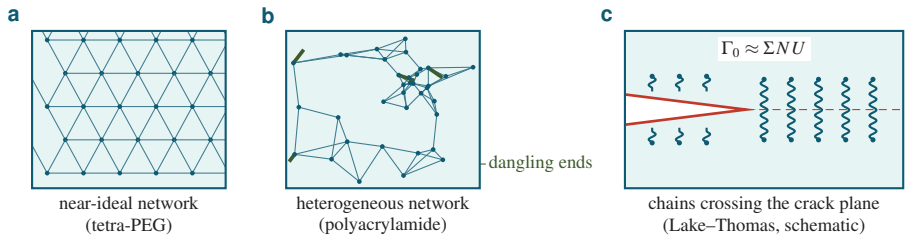


Figure 1. Network architecture and fracture energy (schematic). (a) Tetra-PEG gels approach an ideal, homogeneous network. (b) Conventional polyacrylamide gels contain dense clusters and dangling ends (green). (c) In the Lake–Thomas picture, the intrinsic fracture energy Γ_0 is set by the areal density Σ of chains crossing the crack plane, the number N of monomers per chain, and the energy U to break a monomer; the project will test how network architecture modifies this estimate and the dissipation around it.

Context. This position is part of the SNSF project "The Hierarchy of Scales in the Fracture of Soft and Slender Bodies" (2025–2029). Whether and how a soft solid breaks depends on several length scales at once: the size of the body itself (its thickness, or slenderness), the nonlinear-elastic zone near the crack tip where the material is stretched far beyond linear response, and the region over which energy is dissipated by molecular processes. In soft materials these scales can be comparable, so classical fracture theories, which rely on their strict separation, break down. The project addresses each scale in a dedicated work package (WP) and connects them through combined experiments, simulations, and theory: experiments at Aix-Marseille Université, simulations in the group of Prof. D. Kammer at ETH Zurich, and analytical theory with Prof. M. Adda-Bedia (CNRS, Lyon). The experimental program is carried out at IUSTI (UMR 7343, CNRS / AMU) in the group of J. Kolinski, equipped with custom tensile loading devices, 3D light-sheet microscopy, and particle-tracking infrastructure, with access to AMU's cryo-electron microscopy facilities. IUSTI and the wider Institut Mécanique et Ingénierie (IMI) have strong expertise in rheology, granular and suspension mechanics, and solid mechanics; collaborations within the lab and across the IMI are strongly encouraged, as are research stays at ETH Zurich. The project will host an international workshop on soft fracture at Monte Verità (Ascona, Switzerland) in November 2027.

Candidate profile. We seek a highly motivated candidate holding, or about to obtain, a Master's degree (M2 or equivalent) in physics, physical chemistry, materials science, mechanics, or a related field, with a taste for experiments and quantitative analysis. Experience with polymer or hydrogel synthesis, mechanical testing, or electron microscopy is welcome but not required; training will be provided. The PhD will be carried out on a three-year doctoral contract at Aix-Marseille Université, supervised by J. Kolinski, with day-to-day support from the project's postdoctoral researcher and opportunities for research stays at ETH Zurich. The working language of the group is English.

Contact and application. Send a CV, a one-page statement of interest, and the names of two referees to John Kolinski, john.kolinski@univ-amu.fr • kolinski.science. Applications will be reviewed until the position is filled.

[1] G. J. Lake and A. G. Thomas, "The strength of highly elastic materials," *Proc. R. Soc. Lond. A* **300**, 108–119 (1967).
[2] Y. Akagi *et al.*, "Fracture energy of polymer gels with controlled network structures," *J. Chem. Phys.* **139**, 144905 (2013).
[3] T. Sakai *et al.*, "Design and fabrication of a high-strength hydrogel with ideally homogeneous network structure from tetrahedron-like macromonomers," *Macromolecules* **41**, 5379–5384 (2008).
[4] R. S. Rivlin and A. G. Thomas, "Rupture of rubber. I. Characteristic energy for tearing," *J. Polym. Sci.* **10**, 291–318 (1953).
[5] J. R. Rice, "A path independent integral and the approximate analysis of strain concentration by notches and cracks," *J. Appl. Mech.* **35**, 379–386 (1968).